

David Cai

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EDUCATION

University of Texas at Arlington

Bachelor of Science in Computer Science

Arlington, TX

Dec 2027

University of California, Irvine

Full Stack Development Certificate

Irvine, CA

Oct 2018 – Apr 2019

EXPERIENCE

Software Engineer Intern, Technology Internship Program

Jun 2026 – Aug 2026

Capital One

Plano, TX

- Built and shipped an AI summarization pipeline in Python/JupyterHub that ingests live loan application data from an internal API, preprocesses it, and generates summaries explaining why a case remains pending, replacing manual document review; adopted by the auto loan funding team and rolling out to nearly 100 users.
- Implemented retrieval-augmented generation (RAG) using a Chroma vector store and OpenAI embeddings to ground gpt-oss-120b summaries in source documents, sharply reducing the hallucinations that made early output unusable.
- Proposed and led a user feedback loop, running recurring accuracy surveys and live demos with end users and iterating on prompts and retrieval over several weeks to reach an 80% user-rated accuracy rate.
- Cut per-case context gathering from 4 minutes to 2 minutes, partnering weekly with a project manager, data scientist, and UI/UX designer to align goals and prioritize iterations.
- Shipped the prototype to a runtime environment after management review and handed it off as the foundation for the next team to build on.

PROJECTS

Egg Basket | *Python, LangGraph, PostgreSQL, React/TypeScript, Vercel*

Jun 2026 – Present

- Designed a hierarchical multi-agent architecture in Python/LangGraph for an autonomous investing research system: three Claude Haiku workers evaluate independent signals in parallel and report to a Claude Sonnet supervisor that reconciles them into one allocation, reserving the stronger model for the judgment step.
- Constrained the LLM to narration rather than structure by routing every decision through a deterministic, code-only guardrail layer (25% position cap, 5% cash floor, proportional scale-down), so agent output cannot violate an invariant regardless of what the model returns.
- Split decision from execution across two phases: agents emit dimensionless target weights that a deterministic executor later resolves into concrete orders, keeping model output reproducible and decoupled from live market prices.
- Made unattended daily runs safe to retry using idempotent all-or-nothing writes and an automated post-run integrity check, orchestrated on GitHub Actions.
- Modeled an auditable data layer in PostgreSQL/SQLAlchemy over four external market APIs, and shipped a React 19 + TypeScript dashboard on Vercel publishing every agent decision and its reasoning.

TECHNICAL SKILLS

Languages: JavaScript/TypeScript, HTML/CSS, Python, C, Java, SQL

Frameworks: React, Express.js, LangGraph, SQLAlchemy

AI/ML & Data: Retrieval-Augmented Generation (RAG), Anthropic Claude API, OpenAI API, Chroma, pandas, JupyterHub

Developer Tools: Git, GitHub, GitHub Actions, Vercel, Supabase, Google Cloud Platform, VS Code, IntelliJ, Eclipse, PostgreSQL, MongoDB, RESTful APIs, Node.js

Libraries: Redux, TanStack Query, Recharts, Tailwind CSS, shadcn/ui, Mantine UI, Chakra UI

STUDENT ORGANIZATION

Treasurer - Mobi Byte (Web Development Club)

Jan 2025 – Present

University of Texas at Arlington

Arlington, TX

- Responsible for managing club finances, including taxes, budgeting, funding allocations, purchase approvals, and financial audits.
- Active engagement with overall club strategy, operations, and event planning as part of the leadership team.
- Mentoring new members on development practices and tools, fostering an inclusive learning environment.
- Design and lead technical workshops on web development topics to support member growth and engagement.